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Surrogates for Monte Carlo uncertainty propagation with twenty model runs

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***Abstract*— Monte Carlo propagation of distributions needs far more model evaluations than an expensive simulation allows, so the model is replaced by a surrogate trained on about twenty runs. We compare surrogate strategies on four test models, including a Gaussian plume model of an ammonia release, over 30 paired repetitions, scoring propagated quantiles and exceedance probabilities against reference Monte Carlo values. The kernel specification of a Gaussian process changed the error up to eighteen-fold; one hyperparameter held fixed produced an apparent failure that an earlier analysis attributed to domain size. Logarithmic transformation of positive variables reduced the error up to eightfold. Committees of surrogates beat a well-specified Gaussian process only where one member matched the response structure, and only with error weighting; the adaptive designs tested gave no consistent gain. Cross-validation estimated the surrogate error conservatively; the Gaussian process's own predictive uncertainty understated it up to threefold.**

***Keywords*—Measurement uncertainty; Monte Carlo method; Surrogate model; Gaussian process; Computer experiments**

I. INTRODUCTION

Evaluating measurement uncertainty by the propagation of distributions, following Supplement 1 to the Guide to the Expression of Uncertainty in Measurement (GUM S1) [1], [2], requires the measurement model $Y = f(\mathbf{X})$ to be evaluated once per Monte Carlo trial, typically 10^5 to 10^6 times [3], [4]. For many quantities of practical interest, f is itself a numerical simulation. One example is the concentration downstream of an accidental ammonia release [5]: a single computational fluid dynamics (CFD) run takes hours to days, and a practical budget is often fewer than twenty runs.

The usual remedy is a *surrogate* (emulator) trained on those runs and used in place of f inside the Monte Carlo loop [6]–[8]. The surrogate's error then becomes a contribution to the uncertainty budget [9]. Gaussian process (GP) regression [10], [11] is the standard surrogate; a common guideline asks for about ten runs per input [12], which many applications cannot afford, and results are known to depend on modelling choices [13]. Several alternatives have been proposed for small budgets: weighted committees of different surrogates [14], [15], adaptive selection of runs [16] and, recently, spectral measures of disagreement [17].

An earlier analysis of such a scheme (heterogeneous committee, spectral acquisition, dynamic model management) reported that a passively sampled GP fails catastrophically on large input domains and that the committee recovers accuracy. This contribution tests which of those conclusions survive a controlled comparison with correctly specified baselines, scores the strategies on the quantities a GUM S1 evaluation reports, and asks how far the surrogate's own error can be estimated from the training runs alone.

Table I. Test models; d: number of inputs.

Model	d	Inputs; output
Physics	3	$[-200, 200]^3$
Plume	5	release, wind, height, distance, offset; concentration in mg/m^3
Borehole	8	standard ranges; flow in m^3/yr
Ishigami	3	$[-\pi, \pi]^3$

II. PROPOSED CONTRIBUTION

A. Test models

Four models with uniform input distributions are used (Table I). *Physics* is the benchmark of the earlier analysis; its dominant quadratic term lies exactly in the class of one committee member. *Plume* is the ground-level concentration of a continuous ammonia release from a Gaussian plume model with Briggs rural dispersion coefficients for neutral stability and ground reflection. Its output spans three orders of magnitude and has a 12 % probability of exceeding the level immediately dangerous to life or health (IDLH) for ammonia, 210 mg/m^3 (about 300 ppm). The plume model is an analytical, controlled proxy for dispersion problems, cheap enough for exact reference values; it is not a CFD simulation, and CFD responses with recirculation or regime changes may behave differently. *Borehole* (water flow) and *Ishigami* are standard test functions for computer experiments. No committee member matches the last three.

B. Strategies

Every strategy uses $N = 20$ runs of the model. The single-GP strategies use a 20-point Latin hypercube with five kernel

specifications: the original kernel of the earlier analysis, a squared exponential with one length scale whose amplitude is *fixed* at 1 on the standardised output (not estimated); with the amplitude estimated; with one length scale per input (automatic relevance determination, ARD); with both; and a Matérn-5/2 ARD kernel with amplitude. Hyperparameters are estimated by maximising the log marginal likelihood (L-BFGS-B, five random restarts) on standardised inputs and outputs, with length scales in $[10^{-2}, 10^{3}]$, amplitude in $[10^{-3}, 10^{3}]$ and a nugget variance in $[10^{-8}, 10^{-1}]$. A further variant fits the Matérn GP to the logarithm of the output, with logarithmic transformation of inputs whose range is strictly positive (plume and borehole only); its predictions are back-transformed as the exponential of the latent mean, i.e. the predictive median, without bias correction. The adaptive GP starts from five Latin hypercube runs and adds 15 at its largest posterior standard deviation [16].

The committee has nine members: two GPs (with amplitude and ARD), polynomial chaos expansions (PCE) [18], [19] of degrees 2 and 3, two random forests, a support vector regressor and two neural networks. Its members are combined either uniformly or with weights

$$w_k \propto \left[\mathrm{MSE}_{\mathrm{CV},k}\,(1+\pi_k)\right]^{-1}, \qquad (1)$$

where $\mathrm{MSE}_{\mathrm{CV},k}$ is the three-fold cross-validation error of member k and $\pi_k \geq 0$ penalises predictions below a known physical bound [14], [15]. The committee is trained either on the 20-point design (passive) or on five runs plus 15 chosen where its weighted members disagree most (adaptive). Disagreement is scored by the variance of the member predictions or by UEigV, the eigenvalue-based measure that Lin et al. [17] derived from the normalised graph Laplacian of pairwise similarities between sampled language-model outputs; here the graph is built from the K member predictions at a candidate input.

C. Evaluation

Each strategy is run with 30 seeds. Seeds are shared, so passive strategies use the identical design and adaptive strategies the identical start. The trained surrogate propagates $3 \cdot 10^4$ input draws, and the propagated mean, standard deviation (SD) and 95th percentile $y_{0.95}$ are compared with a reference Monte Carlo evaluation of the true model ($M = 2 \cdot 10^5$). Two exceedance assessments are distinguished: $P(Y > y_{0.95})$ at the reference 95th percentile, whose true value is 0.05 by construction and which tests tail accuracy on every model; and, for the plume, $P(C > \mathrm{IDLH})$ at a fixed physical threshold, which is the quantity a risk decision uses. Accuracy over the input domain is the root mean square error (RMSE) on 5000 test points divided by the output SD (normalised RMSE; 1 corresponds to a constant predictor). Strategies are compared by two-sided Wilcoxon signed-rank tests on paired log normalised RMSE; the 36 tests reported are treated as exploratory, and p-values are Holm-adjusted for multiplicity. All of these errors are *validation* errors that need the reference model; Section D examines what can be estimated without it. Code and every per-seed record are openly available [20].

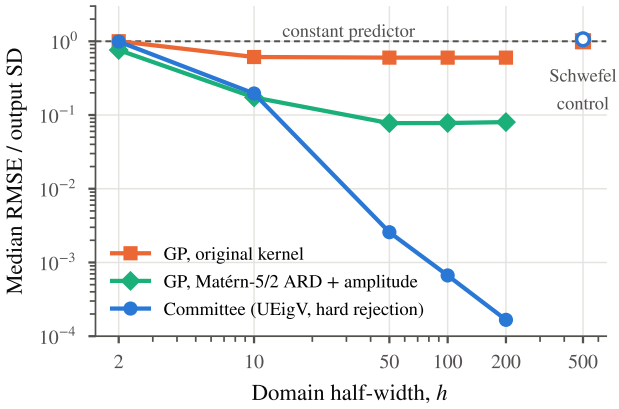


Fig. 1. Physics model on $[-h, h]^3$: median normalised RMSE over 30 seeds against the domain half-width. Open markers: Schwefel function on $[-500, 500]^3$. Dashed line: constant predictor.

Table II. Median normalised RMSE of a single GP on a 20-point Latin hypercube, 30 seeds. Amp.: amplitude estimated; Matérn: Matérn-5/2 kernel.

Kernel	Phys.	Plume	Boreh.	Ishig.
Original (amp. fixed)	0.591	0.602	0.295	0.956
+ amp.	0.033	0.590	0.247	0.970
+ ARD	0.608	0.632	0.129	1.01
+ amp. + ARD	0.036	0.596	0.085	0.896
Matérn, amp. + ARD	0.079	0.597	0.089	0.902
on log variables	–	0.418	0.011	–

III. PRELIMINARY RESULTS AND POTENTIAL APPLICATIONS

A. Kernel specification and variable transformation

Table II shows that the specification of a single GP matters more than any other choice examined. On the physics model the original kernel reaches 0.59, close to a constant predictor; estimating the amplitude alone reduces this to 0.033 (all 30 seeds improve, adjusted $p < 10^{-6}$). The catastrophic GP failure reported earlier is therefore a specification artefact (Fig. 1): with the amplitude fixed, the normalised RMSE is 0.60 at every domain half-width from 10 to 200, whereas a Matérn-5/2 ARD GP with amplitude reaches 0.17 at 10 and 0.08 from 50 to 200. The committee remains far more accurate on this model, but because of its degree-2 PCE member, not because the GP fails. On the borehole model ARD matters (0.30 to 0.09). Logarithmic transformation then gives the largest single improvement: from 0.089 to 0.011 on the borehole model and from 0.60 to 0.42 on the plume (30 and 27 of 30 seeds improve; adjusted $p < 10^{-6}$ and $p < 10^{-3}$). No GP does better than 0.9 on the Ishigami function, for which the ten-runs-per-input guideline [12] would ask for 30 runs.

Table III. Propagation validation error against the reference Monte Carlo evaluation, median over 30 seeds: relative error (%) of the mean and of $y_{0.95}$, and absolute error (×100) of $P(Y > y_{0.95}) = 0.05$. Bold: best per column.

	Physics			Plume			Borehole			Ishigami		
Strategy	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P	mean	$y_{0.95}$	P
GP, original kernel	8.90	21.1	4.86	10.4	20.4	3.29	1.60	13.0	4.25	10.3	34.6	4.73
GP, Matérn-5/2 ARD	1.36	3.88	1.37	9.68	17.7	2.87	0.66	2.45	0.91	**8.48**	**16.0**	3.61
GP, log variables	–	–	–	**4.74**	**6.94**	**0.76**	**0.17**	**0.60**	**0.18**	–	–	–
Committee, CV-weighted	**0.21**	**0.41**	**0.15**	8.75	23.1	3.15	1.06	10.9	3.05	11.0	28.7	**2.89**
Committee, adaptive	**0.21**	**0.41**	**0.15**	55.1	20.5	5.39	3.30	1.41	0.49	15.9	34.9	4.40

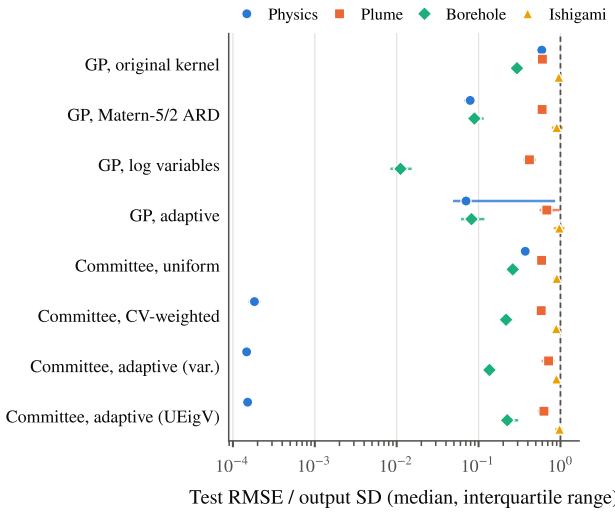


Fig. 2. Normalised RMSE of each strategy on each test model: median (marker) and interquartile range (bar) over 30 seeds; logarithmic axis. Dashed line: constant predictor. Committees use GP members with amplitude and ARD.

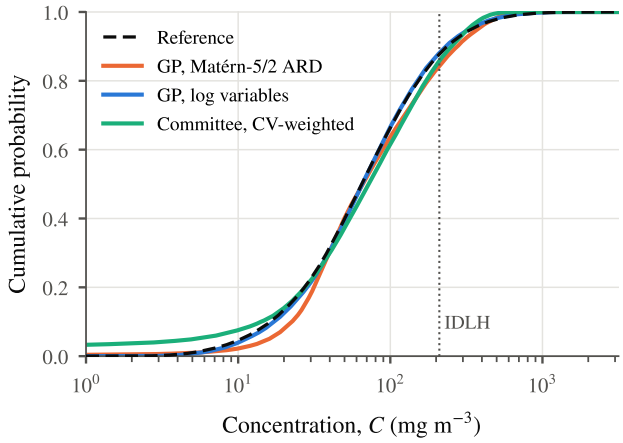


Fig. 3. Plume model, first seed: distribution of the ground-level concentration propagated through three surrogates fitted to the same 20 runs, and the reference Monte Carlo distribution. Dotted: IDLH level for ammonia.

B. Committees and adaptive runs

Fig. 2 compares all strategies. A committee outperforms the best single GP only on the physics model, by more than two orders of magnitude, and only because the degree-2 PCE represents its dominant term exactly; the weights then concentrate on that member. On the borehole model the CV-weighted committee is 2.3 times worse than the Matérn-5/2 GP on the same design (0 of 30 seeds better), and on the plume and Ishigami models the two tie. Uniform averaging was worse than CV weighting on the physics (by a factor of about 2000) and borehole models (adjusted $p \leq 0.002$) and never better. Within the designs tested (five initial runs, 15 adaptive runs chosen by posterior SD, committee variance or UEigV), adaptive selection gave no consistent benefit. For the single GP it was never significantly better. For the committee it helped on the physics and borehole models (median ratios 0.80 and 0.61), hurt on the plume (1.25, adjusted $p = 0.005$) and made no difference on the Ishigami function. UEigV was never better than variance acquisition after adjustment, and worse on the borehole model by a factor of 1.7 (adjusted $p < 10^{-6}$). Acquisition functions aimed at a specific quantile or threshold may behave differently.

C. Propagated quantities

Table III gives the validation errors of the quantities a GUM S1 evaluation reports, and they follow the normalised RMSE. With the best strategy for each model, the 95th percentile is within 0.41 % (physics), 6.9 % (plume), 0.60 % (borehole) and 16 % (Ishigami) of the reference, and $P(Y > y_{0.95})$ within 0.0015, 0.0076, 0.0018 and 0.029 of 0.05. At the fixed IDLH threshold (reference $P(C > \text{IDLH}) = 0.123$), the log-variable GP has a median absolute error of 0.009 over 30 seeds and the raw-variable GP 0.023; Fig. 3 shows one seed. Adaptive runs can mislead: the adaptive committee estimated the plume mean with a median error of 55 %.

D. Estimating the surrogate contribution

In a real evaluation the reference is unavailable, and the surrogate's contribution must be estimated from the twenty runs. Table IV compares two estimates with the true error for the recommended GPs: the five-fold cross-validation RMSE (hyperparameters re-fitted in every fold) and the root mean square of the GP's own predictive SD. Cross-validation was conservative: it was at least the true error in 28 and 30 of 30 seeds on the physics and borehole models, close to it on the plume and Ishigami models, and below half of it in at most 6 of 30 seeds (plume, log variables). The GP's own uncertainty was unreliable in the unsafe direction

Table IV. Estimating the surrogate error from the 20 runs, median over 30 seeds: ratio of the five-fold cross-validation RMSE and of the GP's root-mean-square predictive SD to the true RMSE, and coverage of the nominal 95 % predictive interval. M: Matérn GP; L: on log variables.

Model	GP	CV / true	SD / true	Coverage
Physics	M	9.06	1.58	100 %
Plume	M	0.99	0.36	77 %
Plume	L	0.81	0.42	73 %
Borehole	M	2.11	0.79	90 %
Borehole	L	2.61	1.45	99 %
Ishigami	M	1.07	0.58	73 %

where the GP is weakest: on the plume and Ishigami models it understated the error by a factor of 1.7 to 2.8, and its nominal 95 % predictive intervals covered 73 to 77 % of test points. The member envelope of a committee was no better a substitute: read as an interval for $P(Y > y_{0.95})$, it contained the reference value in 0 to 100 % of seeds depending on model and strategy [21].

These results suggest a simple, conservative treatment for a GUM S1 evaluation: propagate $Y = \hat{f}(\mathbf{X}) + E_\mathrm{s}$, where $\hat{f}$ is the surrogate and the surrogate error E_s is an additional input with expectation zero and standard uncertainty equal to the cross-validation RMSE. This ignores the spatial variation and correlation of the surrogate error and inflates the uncertainty when the surrogate is accurate; its calibration on a CFD case, for example by conformal prediction [22], is the main task for the final paper.

E. Potential applications

For surrogate-based GUM S1 evaluations with about twenty runs, such as a dispersion study in a risk assessment, the results suggest five rules. *Specify the GP fully*: estimate an amplitude and one length scale per input. *Transform the variables* when the model is multiplicative or positive, for example concentrations and flows. *Use a committee only with error weighting*, and expect a gain only when a member can represent the dominant structure. *Prefer a space-filling design* unless an adaptive criterion has been shown to help for the quantity of interest. *Estimate the surrogate contribution by cross-validation* and include it in the budget, rather than relying on the GP's own predictive variance or on committee spread.

IV. CONCLUSIONS

For Monte Carlo uncertainty propagation with twenty runs of an expensive model, the kernel specification of a GP surrogate and the transformation of the variables mattered more than the choice between single surrogates, committees and the adaptive designs tested. A failure of GP surrogates reported in an earlier analysis was caused by one hyperparameter held fixed. Committees helped only when a member matched the response structure and only with cross-validation error weighting. Cross-validation gave a conservative estimate of the surrogate's error, whereas the GP's own uncertainty could not be trusted. These findings rest on analytical models; the final paper will add a CFD dispersion case, budgets other than twenty runs, and a calibrated treatment of the surrogate contribution.

AUTHORS' CONTRIBUTION

Yugam Jinesh Vora: conceptualisation, methodology, software, investigation, formal analysis, visualisation, writing – original draft. Wee Hoe Ng: conceptualisation, supervision, writing – review and editing.

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